Université d'Ottawa · University of Ottawa

Faculté des sciences
Mathématiques et de statistique

Faculty of Science
Mathematics and Statistics

MAT 2143 – The final exam **Instructor: K. Zaynullin**

Last name: _____

First name: _____

Signature: _____

Student number:

Read the following instructions:

- Not allowed: textbooks, course notes, electronic devices (telephones, computers).
- Do not detach the pages of this examination.
- You may use the back of the pages as scrap paper for calculations, or to answer questions if you run out of space on the front side.
- You must give clear and complete solutions, with calculations, explanations and justifications. Make sure that your answer is clearly indicated.

THIS SPACE IS RESERVED FOR THE MARKER:

Question	1	2	3	4	5	6	7	8	9	10	Total
Mark											
Out of	7	5	5	4	6	6	6	5	5	6	55

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1. Let $n = 54$ and $m = 63$.

(a) Find $gcd(n, m)$. (1)

(b) Express $gcd(n, m)$ as a linear combination of n and m. (2)

(c) Find $lcm(n, m)$. (1)

(d) Determine whether the following statement is true of false (1)

$$n \equiv m \ (mod\ 7).$$

(e) Find all integers $k \geq 2$ such that $n \equiv m \ (mod\ k)$. (2)

2. Let

$$\sigma = \begin{pmatrix} 1\,2\,3\,4\,5\,6\,7\,8\,9 \\ 4\,5\,6\,3\,7\,1\,9\,8\,2 \end{pmatrix}$$

(a) Factor σ into a product of disjoint cycles. (2)

(b) Factor σ^{-1} into a product of disjoint cycles. (1)

(c) Factor σ into a product of cycles of length 2 (transpositions). (1)

(d) Find the parity of σ. (1)

3. Let G be an abelian group.

Prove that the subset $H = \{g^2 \mid g \in G\}$ is a subgroup of G. (5)

4. Let a and b be elements in a group. Prove that
if $a^4 = 1$ and $ab = ba^2$, then $a = 1$. (4)

5. Let G be a group. Define a map $f\colon G \to G$ by $f(g) = g^{-1}$ for all $g \in G$. Prove that G is abelian if and only if f is a homomorphism. (6)

6. Let $G = \mathbb{Z}/8\mathbb{Z}$ be a group with respect to the addition.

Find the right cosets of the subgroup $2\mathbb{Z}$ in G. (6)

7. Let K be a normal subgroup of G. Assume that K is cyclic.

Prove that every subgroup of K is normal in G. (6)

8. Let G be a cyclic group of order 24 and let $H = \langle a^{12} \rangle$ be its subgroup.
Find the order of the element (coset) Ha^2 in the factor group G/H. (5)

9. Given elements r and s in a ring R, prove that

$1 + rs$ is a unit (invertible) in R if and only if $1 + sr$ is a unit (invertible) in R. (5)

10. Let R be a ring. Given an integer m, prove that the subset

$$R_m = \{r \in R \mid mr = 0\}$$

is an ideal of R. (6)

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